JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 104, NO. B4, PAGES 7577–7598, APRIL 10, 1999

Paleointensity variations across the last geomagnetic reversal at La Palma, Canary Islands, Spain

Jean-Pierre Valet, [1]Jacques Brassart,[1] Xavier Quidelleur,[2] Vicente Soler,[3] Pierre-Yves Gillot,[2] and Lionel Hongre[1]

Abstract. A detailed volcanic record of absolute paleointensity has been obtained for the period encompassing the last geomagnetic reversal, from a succession of 69 distinct lava flows in two adjacent sections on the island of La Palma. The correlation between the two sections relies on the positions of the Brunhes-Matuyama boundary and on the flow chronology obtained by a refined potassium/argon (K/Ar) dating technique. Several pretransitional flows exhibit very shallow paleomagetic inclinations, but they cannot be considered as really transitional. Paleointensity experiments conducted on 288 specimens provided 168 successful determinations (14 were obtained after correction) for 50 lava flows that were considered to represent 45 distinct units. The evolution of the paleofield intensity can be constructed by assuming uniform extrusion rates between three dated lava flows. The period preceding the transition is characterized by large fluctuations with typical peak to peak amplitudes between 20 and 40 μT which decrease down to 7 μT prior to the transition. The 100 kyr- long interval following the reversal is associated with field intensities as high as 60 μT and depict a strong and rapid restoration of the dipole field. The virtual dipole moments (VDMs) have been compared with sedimentary VDMs generated from the record of relative paleointensity from Ocean Drilling Program site 851, to which have been added non dipole components similar to the present geomagnetic field. A reasonable fit can be made between the two curves which implies variations in the eruption rates but not incompatible with the uncertainties in the K-Ar age determinations. Thus these results can support but do not fully demonstrate the existence of asymmerical saw-toothed variations in field intensity.

1. Introduction

A major step has been achieved during the past decade with the acquisition of high-resolution sedimentary records of relative paleointensity which document long-term variations of the geomagnetic field. However, the complex processes associated with acquisition of magnetization in sediments cannot be duplicated by laboratory experiments so that sedimentary records are often regarded with limited confidence. The agreement between the composite curve of relative paleointensity (*Guyodo and Valet*, 1996) and the variations in beryllium 10 (*Frank et al.*, 1997) production during the past 200 kyr has recently demonstrated that magnetization of sediments can provide a coherent and reliable description of the geomagnetic changes. If we deal with older periods, one of the most conspicious features observed so far is the saw-toothed shape of field intensity changes across reversals as observed in several records from the Pacific and the Indian oceans (*Valet and Meynadier*, 1993; *Meynadier et al.*, 1994; *Thibal et al.*, 1995; *Verosub et al.*, 1996). However, there are also records which do not exhibit such asymmetry between pre reversal and post reversal periods (*Laj et al.*, 1996). Alternative nongeomagnetic explanations have been proposed (*Meynadier and Valet*, 1995; *Mazaud*, 1996; *Kok and Tauxe*, 1996) for the saw-toothed shape of long-term paleointensity changes but they are based also on assumptions that are controversial (*Meynadier and Valet*, 1996; *Meynadier et al.*, 1998). Determinations of absolute paleointensity from volcanic rocks should help clarify these problems because it is relatively easy to duplicate a thermoremanent magnetization in laboratory. Unfortunately, there are many problems caused by mineralogical changes during heating which frequently yield rejection of a large number of samples. Another fundamental requirement is the need for accurate dating. To date four records of geomagnetic reversals from volcanic sequences (*Prévot et al.*, 1985; *Chauvin et al.*, 1990; *Bogue and Paul*, 1993; *Quidelleur and Valet*, 1996) include determinations of absolute paleointensity. In all the cases the field recovery following the transition is characterized by higher intensities than the period preceding the directional changes. However, these results lack detailed control in terms of chronology so that only time independent observations can be made.

The initial aim of this study was to retrace the variations of the full geomagnetic vector before and after a reversal recorded in a sequence of lava flows. The section of El Time on the island of La Palma incorporates more than 100 flows erupted around the time of the Brunhes-Matuyama (B-M) reversal and was thus quite appropriate to meet these objectives.

2. Volcanics of La Palma and Sampling of the El Time Section

Seven volcanic islands form the Canary archipelago (Figure 1). The western islands lie above oceanic crust, but the central

[1]Institut de Physique du Globe de Paris, UMR 7577 CNRS, Paris.
[2]Laboratoire de Géochronologie, Université Paris XI Orsay and Institut de Physique du Globe de Paris.
[3]Instituto de Productos Naturales y Agrobiologia de Canarias, Eataciòn Volcanològica de Canarias, La Laguana, Tenerife, Spain.

Copyright 1999 by the American Geophysical Union.

Paper number 1998JB900099.
0148-0227/99/1998JB900099$09.00

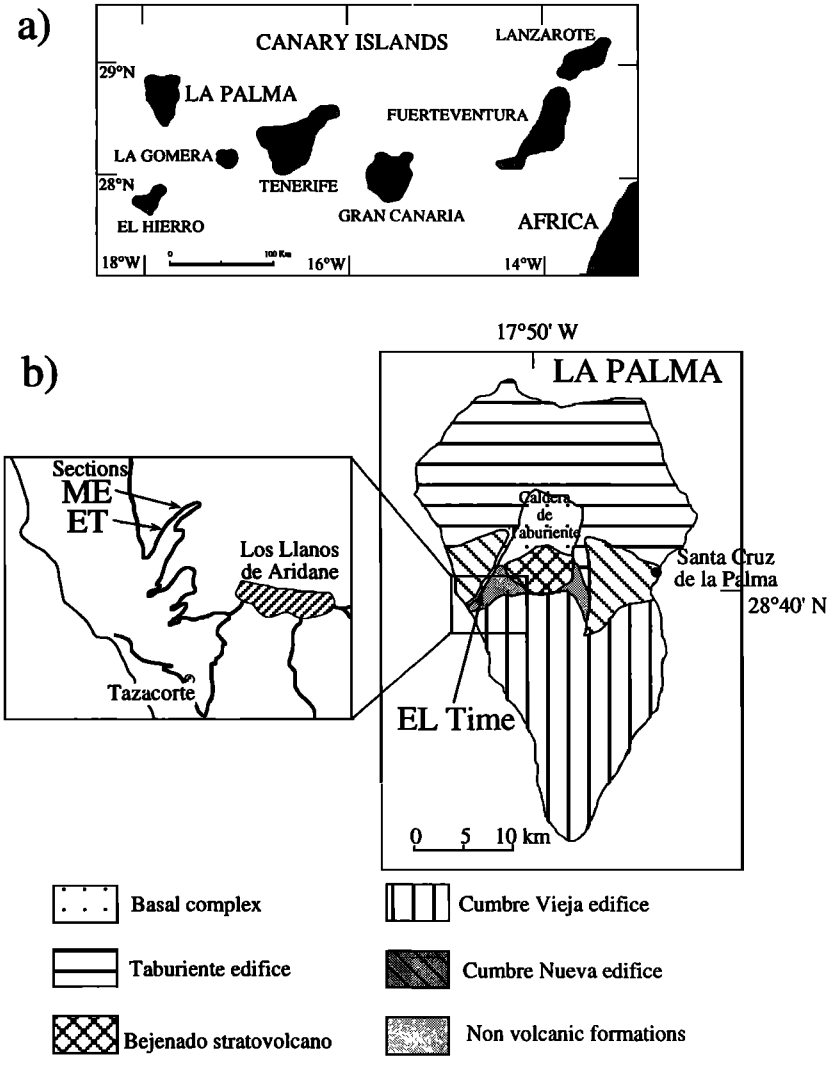


Figure 1. (a) Map of the Canarian Archipelago. (b) Location of the upper El Time (ET) and the main El Time (ME) sections with reference to the main road (left scheme). Schematic geological map showing the location of the El Time sequence.

part of the archipelago is underlain by a transitional zone between oceanic and continental crust (*Dash and Bosshard*, 1969). The studies conducted by *Abdel-Monem et al.* (1972) have shown that volcanism has persisted more or less continuously during the past 20 Myr. It is still active in several islands, particularly in La Palma where the last eruption occurred in 1971 at the Teneguia volcano.

La Palma island, built on Jurassic oceanic crust, rises 6500 m above the seafloor and 2400 m above sea level. Subaerial volcanic activity was initiated 2 Ma ago and built up the northern edifice in two successive phases (*Fuster et al.*, 1989) between 2 and 1.3 Ma (lower old series) and between 1 and 0.7 Ma (upper old series). The most remarkable geological feature of La Palma island is the Caldera de Taburiente (Figure 1) which according to *Ancochea et al.* (1994) has been formed by large landslides. The El Time section is located on the northwestern flank of the Caldera. According to *Ancochea et al.* (1994) the entire section belongs to the upper old series. An alternative view has been proposed by *Navarro and Coello* (1993). According to them the upper part of the section should be incorporated within the Cumbre Nueva Series and thus associated with a different eruptive center (see Figure 1). Two lava flows have been dated by potassium-argon (K-Ar) by *Abdel-Monem et al.* (1972), and the ages have been recalculated by *Ancochea et al.* (1994). Ages of 1.05 Ma and 0.73 Ma for the middle and the top of the section indicate that the upper part of the sequence incorporates the Brunhes-Matuyama reversal.

The paleomagnetic sampling of El Time was confined to the upper 100 flows of the sequence. Sampling strategy was guided by field measurements with a portable fluxgate magnetometer to determine the magnetic polarity and hence the position of the Brunhes-Matuyama reversal. Sampling was achieved during three successive field trips. Particular attention was taken to sample steep sections so that the flow succession could be identified without ambiguity. For these reasons we prefered to sample two distinct sections rather than the apparent succession of flows outcropping along road C-832. Twenty lava flows sampled about 50 m below a "Camino real" that crosses the sequence compose the upper El Time (ET) section, the four lowermost with reverse polarity outcropping along or immediately above road C-832. The main El Time (ME) section is located on the same flank of the mountain, 600 m to the east

of the ET section. A total of 49 lava flows were sampled. The positions of the B-M reversal was initially detected by fluxgate measurements between flows ME44 and ME47 and flows ET04 and ET06. Several sills and dikes have been observed in the ME section which are indicated in Figure 2. The base of the section is characterized by a massive unit (ME01), the origin of which (sill or flow) could not be unambigously determined in the field. The results of new K-Ar dating described below confirmed that this was in fact a sill. The majority of the flows are between 1 and 2 m thick. Given difficulties in following and correlating these thin flows over a long distance, it is very difficult to match up the two sections using only field observations. The ME section is closer to the inner part of the caldera (i.e., closer to the eruptive center) than the ET section and thus incorporates more flows.

A total of 445 cores were taken with a portable gasoline powered drill and oriented with a magnetic compass. The cores were mostly taken in the lower part of the flows to avoid possible reheating effects by the overlying unit. In all cases they were collected over horizontal distances of 10 to 20 m. In the most massive units, cores were drilled at different vertical positions. Whenever possible Sun orientation has primarily been used. The maximum discrepancy between the Sun and the magnetic orientation of cores does not exceed 12° with a mean discrepancy of 6.5° and 3.5° for the ET and ME sections, respectively.

3. Magnetic Carriers and Natural Remanent Magnetization (NRM)

3.1. Magnetic Mineralogy

Thermomagnetic measurements were performed on 10 specimens from different lava flows in air and/or under argon

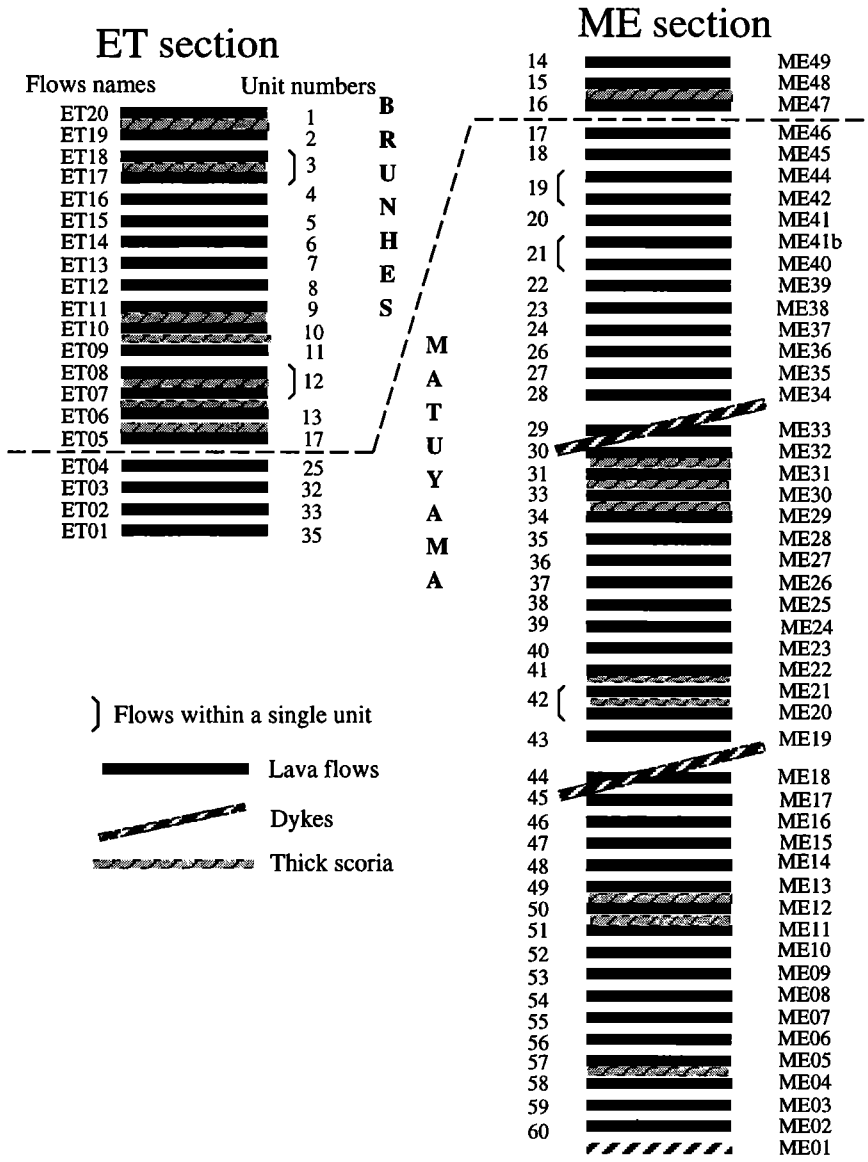


Figure 2. Distribution of the successive lava flows in the ET and ME sections, respectively. The dashed line separates the Matuyama and Brunhes polarity chrons. The external side references give flow identifications. The internal numbers indicate the unit numbers obtained after correlating the successive lava flows with similar directions, paleointensity, and rock magnetic properties (see text and tables) .

atmosphere with the Institut de Physique du Globe (IPG)-Orsay horizontal Curie balance. Very similar results have been observed with both atmospheres. Most thermomagnetic curves can be broadly separated in two different categories. For a large majority of the specimens the saturation magnetization (Js) is very stable to temperatures as high as 450-520°C and then decreases rapidly (Figure 3a). The cooling curves match more or less the heating curves without significant production or destruction of magnetic material. The Curie temperature, between 560°C and 580°C, indicates either pure magnetite or titanomagnetite with a small amount of titanium. The other specimens (Figure 3b) are characterized by a more regular decrease of Js, similar to the decrease of the natural remanent magnetization (NRM) during stepwise demagnetization. In a few specimens, production of magnetic material with blocking temperatures higher than the last heating step was detected by a significant increase of Js on cooling. In such situations the cooling curve differs strongly from the heating curve. The NRM of the companion specimen was affected by production of chemical remanent magnetization (CRM) during heating in presence of a magnetic field; these specimens were therefore unsuitable for paleointensity experiments.

3.2. Rock Magnetic Properties

Before proceeding to paleointensity measurements we have scrutinized several rock magnetic parameters. The viscosity index (v) of 235 specimens has been calculated by using a modification of the original Thellier method [*Thellier and Thellier*, 1944] in which the *z* axes of the specimens were oriented parallel and then antiparallel to the direction of the ambiant magnetic field (40 μT) for 10 days, respectively. The stable NRM is then calculated as half the sum and the viscous remanent magnetization (VRM) as half the difference of the total remanences measured after each cycle. Seventy-seven percent of the specimens are characterized by a v value (v = VRM/NRM x 100) lower than 6%. Seven flows from the ME section have v as high as 40%. No paleointensity determination was obtained in these cases (if we remove these flows, the mean value of v is 4.5). Eighty-five percent of the specimens have a viscosity lower than 10 with a mean value of 2.5.

Differences in magnetic concentration between successive lava flows can be inferred from low field susceptibility (K) and anhysteretic remanent magnetization (ARM). Acquisition of anhysteretic remanent magnetization was done with a

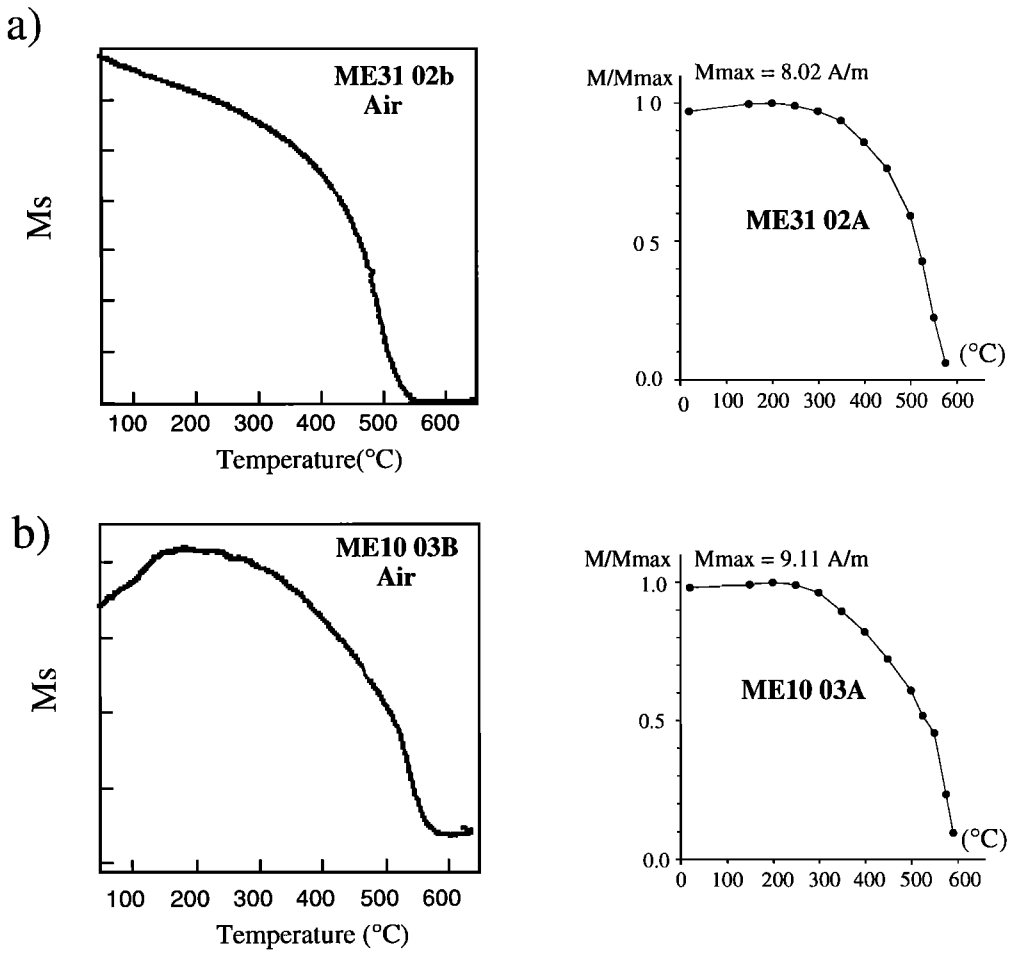


Figure 3. Evolution of magnetization intensity during thermal demagnetization. (right) The unblocking temperatures (Tub) of the NRM are compared with those derived from thermomagnetic experiments in high field (left side). Two different situations are depicted for (a) ME31 02b and 02A and(b) ME10 03B and 03A (see text).

Schonstedt field demagnetizer, and K was measured with a Bartington susceptibility meter. The variations of these two parameters are very similar and do not exceed a factor of 3 (Figure 4a), but their respective amplitudes are different, which could reflect changes in magnetic granulometry. Magnetic granulometry may play a significant role in paleointensity experiments, particularly when multidomain grains are dominant. Two different techniques were tentatively used to determine the range of magnetic grain sizes. Following the approach developed by *King et al.* [1982, 1983], the anhysteretic susceptibility (defined as the ARM obtained in a 0.1mT steady field and 100 mT alternating peak field) has been plotted versus K (Figure 4b). This gives a rough estimate of the variability in magnetic grain sizes. Keeping in mind the limits inherent to this technique, the straight lines emanating from the origin are consistent with single- or pseudo-single-domain grain sizes of magnetite. High field hysteresis measurements have been done with a translation inductometer to a maximum applied field of 1 T in the IPG laboratory of Parc St. Maur. One advantage of this technique is that the measurements can be carried out on large amounts of powder and are thus not affected by inhomogeneities. The results plotted on a Day diagram [*Day et al.*, 1977] in Figure 4c indicate that the grain sizes range in the pseudo-single-domain area with a tendency toward single-domain grains. Note that there is a rather good coherency between hysteresis measurements and indications derived from the ARM/K ratio. The specimens with a low ARM/K ratio were found to lie closer to the multidomain area of the Day diagram.

3.3. Magnetization Components

Measurements were performed in the paleomagnetic laboratory of IPG, using a JR5 spinner magnetometer. The initial measurements (about 60% of the specimens) were performed outside the shielded room while the other 40% were measured inside. Except for a few specimens with high magnetic viscosity we did not notice any significant improvement in the quality of the results after the second set of experiments.

Thermal demagnetization of the NRM was integrated within the procedure of the paleointensity experiments. In most cases, three specimens per lava flow have been thermally demagnetized and at least two additional specimens were af stepwise demagnetized up to 100-110 mT peak fields. In Figure 5 we show af and thermal demagnetization diagrams. Alternating fields were often more efficient at removing the secondary components, in relatively low fields. The stepwise demagnetization of the anhysteretic remanent magnetization (ARM) (in this experiment the ARM was imparted in a 90 mT af peak field and a 0.05 mT steady field) duplicates the characteristics of the NRM (Figure 6). We found more interesting to compare the median destructive fields (mdf) without incorporating the NRM in order to deal with the stable part of the remanence. The mdf values range between 25 mT and 40 mT and do not show large variations between the successive lava flows. These observations suggest that fine grains are associated with the primary magnetization and that there are no very large changes in the magnetic behavior of the successive flows.

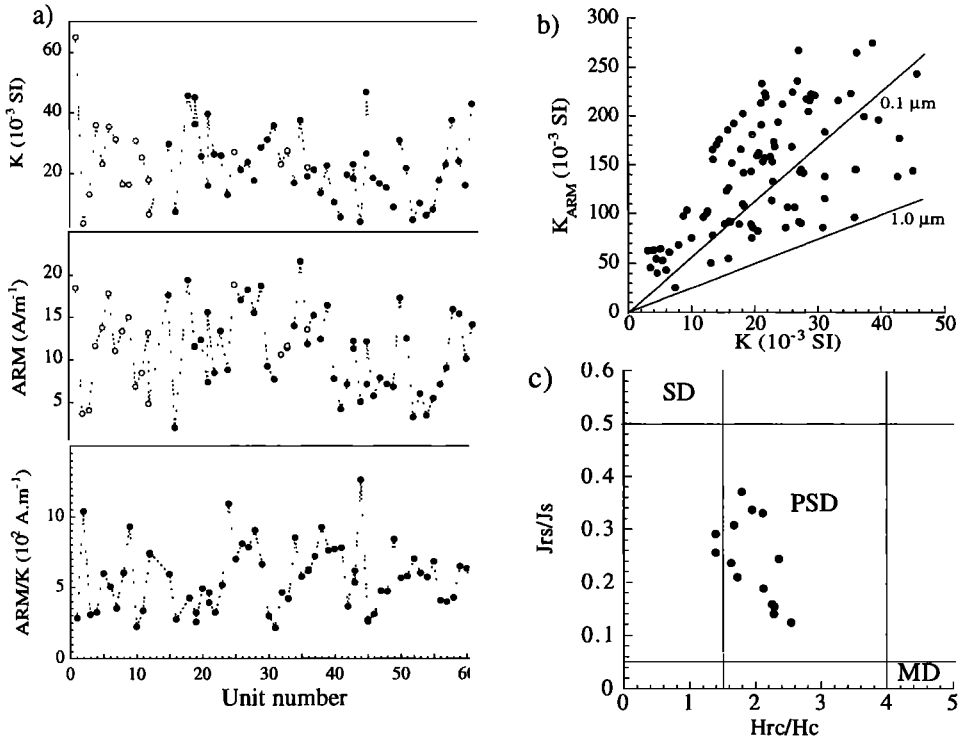


Figure 4. (a) Variations of ARM, K, and ARM/K ratios for the ET (open symbols) and the ME section (solid symbols). Note the existence of similar ARM/K ratios for some successive flows. This parameter was used to define successive volcanic units along with identical directions and paleointensity measurements (this plot includes additional measurements performed from flows ME 42 to ME 47). Range of magnetic grain sizes derived from (b) ARM versus K diagrams and (c) from hysteresis, SD, single domain ; PSD, pseudo-single domain, and MD, multidomain.

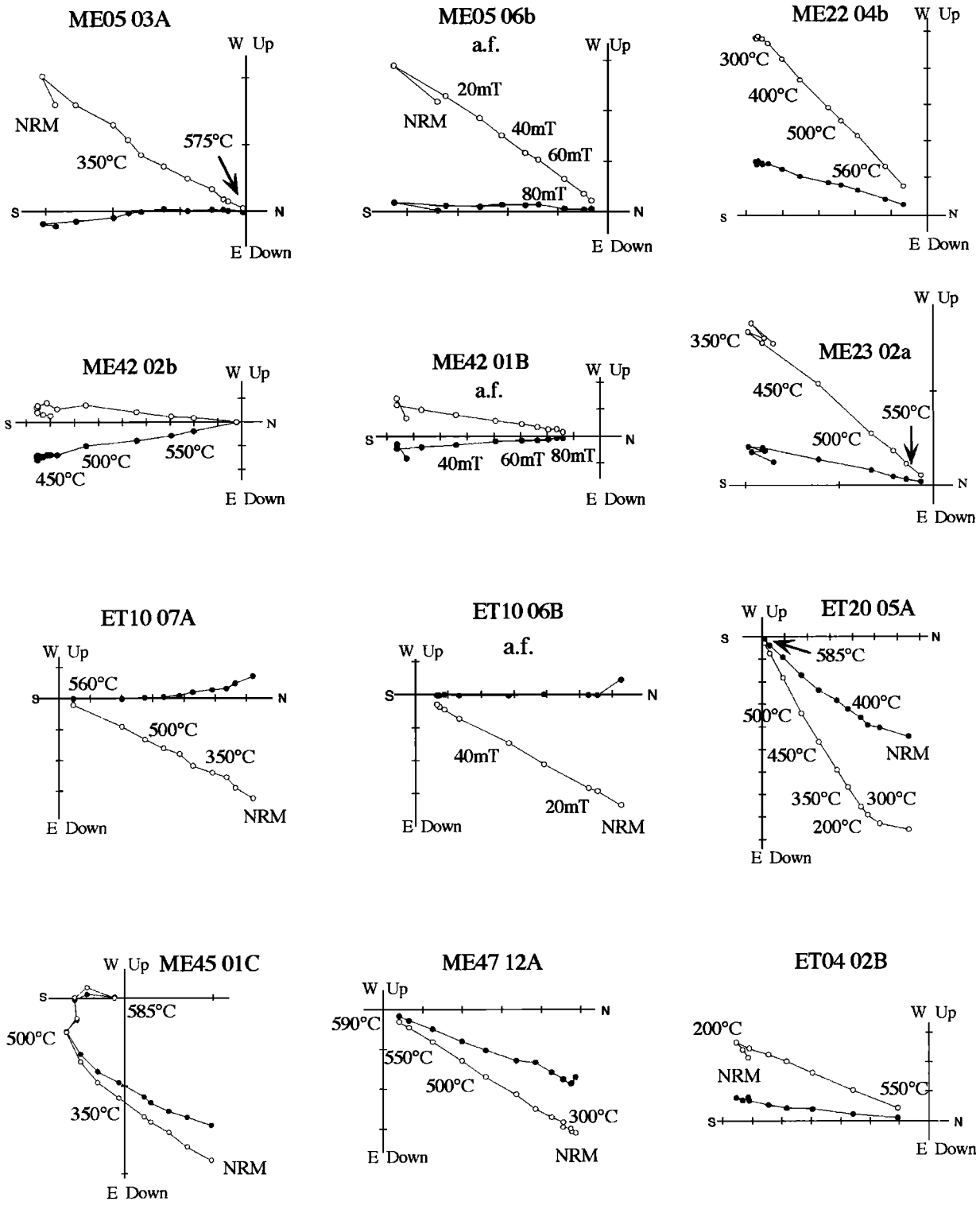
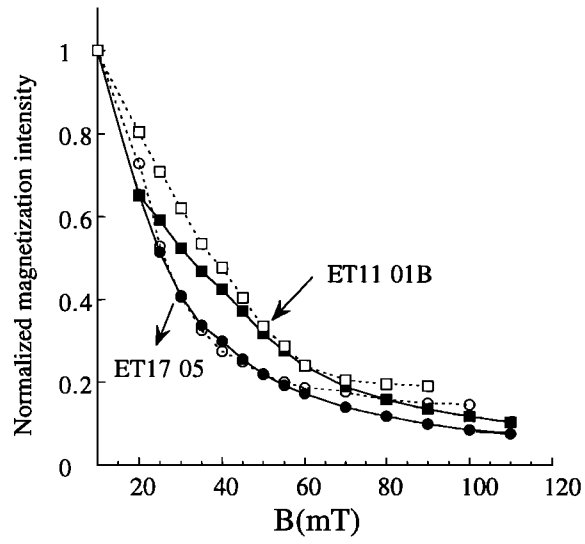
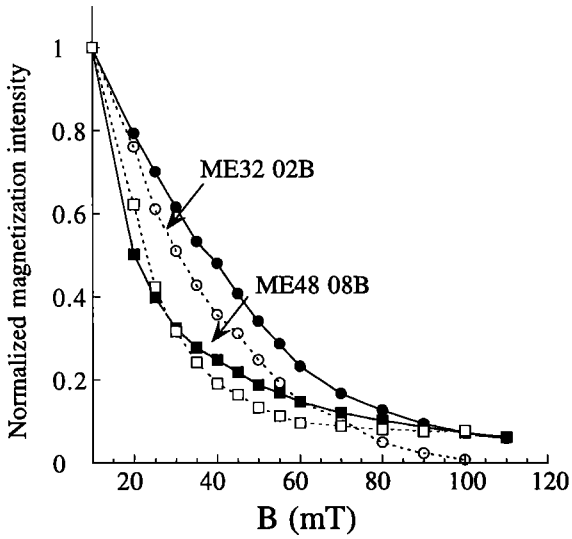
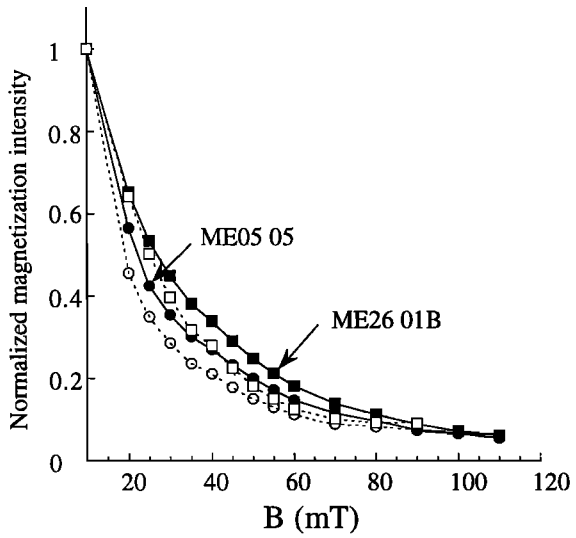


Figure 5. Thermal and alternating field demagnetization diagrams of specimens distributed at different levels in the ET and the ME sections (inclinations are shown by open symbols). Specimen ME45 01c shows atypical behavior with a strong overprint of normal polarity likely caused by thermochemical remagnetization (see text).







Three components of magnetization could be isolated during thermal demagnetization. In a few cases a very soft component with no preferential direction was likely acquired during storage and easily removed at 100°C. A second component with a direction close to the present-day field and thus likely of viscous origin was removed between 200°C and 300°C. Beyond these steps the characteristic high-temperature component decreases during demagnetization up to 550-580°C. The samples from flows ME45 and ME46 (immediately below the first flow with normal polarity) exhibited a distinctive behavior, with a normal polarity component associated with unblocking temperatures higher than 500°C and a final tiny reverse polarity component (Figure 5). The most plausible hypothesis is that the medium-high-temperature (HT) component resulted from thermochemical remagnetization following the emplacement of the overlying flow [*Valet et al.*, 1998]. About 20% of the demagnetization diagrams shows slight curvature between 300°C and 450°C, which could be associated with similar processes. In such cases the directions and paleointensity estimates were determined beyond these temperatures.

The final declinations and inclinations of the primary component of magnetization have been calculated by principal component analysis through the origin of the demagnetization diagrams. The mean flow directions were calculated using *Fisher*'s [1953] statistics. The direction of the HT component of the specimens from flow ME46 was poorly defined and was thus not taken into consideration. The mean directions and their associated VGPs are listed in Tables 1 and 2 for the ET and ME sections, respectively.

4. Determination of Absolute Paleointensity

4.1. Experiments

The modified Thellier double heating method [*Coe*, 1967] has been used for paleointensity experiments. In order to constrain and detect magnetomineralogical changes produced during heating, partial thermoremanent magnetization (pTRM) checks were systematically performed one temperature step below the last pTRM acquisition. The routine involved demagnetization of the NRM between room temperature (T_0) and the heating step (T_i) followed by pTRM acquisition (in a field ranging from 35.6 to 42.6 μT depending on the specimen position along the holder) for the same interval of temperature. Each pTRM check was obtained after demagnetization in zero field between T_0 and T_{i-1}. The temperature increased in steps of 50°C between room temperature and 500°C and every 25-30°C at high temperatures. A total of 288 specimens have been processed (103 and 185 for the ET and ME sections, respectively) in batches of 30 specimens within a Pyrox oven in the shielded room of the IPG paleomagnetic laboratory. Most of the experiments (240 specimens) were conducted in air, but 48 specimens were heated

Figure 6. Stepwise alternating field demagnetization of NRM and ARM intensities of specimens from different flows. The similarity of the curves (plotted beyond 10 mT) depicts the contribution of similar spectrums of magnetic grains to both magnetizations and the absence of large variations between the lava flows.

Table 1. Mean Directions and Associated VGPs for the ET Section

Flow	Unit	*n/N*	*D*, deg	*I*, deg	a_{95}	*k*	VGP Latitude, deg	VGP Longitude, deg
ET20	1	6/6	26.1	53.9	3.8	313.6	67.1	50.8
ET19	2	6/6	29.4	62.0	2.5	712.6	62.3	32.4
ET18	3	6/6	24.2	54.2	3.1	625.3	68.6	49.5
ET17	3	5/6	28.9	55.5	2.4	997.0	64.6	47.8
ET16	4	6/6	11.9	34.6	3.2	451.6	75.5	111.0
ET15	5	5/5	7.1	35.5	3.5	485.0	78.9	124.9
ET14	6	6/6	15.2	34.4	4.0	287.3	73.0	103.9
ET13	7	5/6	14.9	29.1	3.9	380.0	71.0	112.5
ET12	8	5/7	3.1	38.8	4.2	482.7	82.7	138.9
ET11	9	4/6	-8.5	32.2	6.6	197.7	76.4	198.8
ET10	10	8/8	-4.9	28.1	1.3	1901.6	75.6	181.4
ET09	11	4/9	-1.7	22.4	4.2	470.7	72.9	167.8
ET08	12	5/9	14.1	31.9	4.0	365.8	72.8	110.3
ET07	12	6/6	16.9	33.6	4.0	276.3	71.4	102.0
ET06	13	8/8	18.7	27.3	3.2	302.9	67.6	107.5
ET05	17	11/15	-11.3	-4.1	4.0	195.2	57.4	183.4
ET04	25	7/13	181.9	-21.6	3.9	243.8	-72.4	335.9
ET03	32	6/9	183.6	-28.3	3.6	353.6	-76.0	327.6
ET02	33	6/6	180.6	-31.0	3.3	419.0	-78.0	339.3
ET01	35	4/6	190.8	-43.5	9.4	97.4	-79.8	268.4
			Averaged Results Within Units					
ET17/18	3	11/12	26.5	54.8	1.9	653.0	66.7	48.7
ET07/08	12	11/15	15.7	32.8	2.6	307.9	72.0	105.7

n/N, number of samples used in the analysis/total number of samples treated; *D*, declination; *I*, inclination ; α_{95}, 95% confidence cone of the Fisher's (1953) statistics; *k*, precision parameter of the Fisher's (1953) statistics.

Table 2 : Mean Directions and Associated VGPs for the ME section

Flow	Unit	*n/N*	*D*, deg	*I*, deg	α_{95}	*k*	VGP Latitude, deg	VGP Longitude, deg
ME49	14	2/2	14.7	29.3	n.d.	n.d.	71.2	112.7
ME48	15	6/8	17.8	30.9	2.5	721.3	69.7	104.5
ME47	16	9/9	15.2	31.6	2.7	367.4	71.9	108.4
ME45	18	7/22	170.5	-3.0	2.0	917.5	-61.4	2.2
ME46	18	0	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
ME42	19	4/4	172.6	-5.5	5.1	331.9	-63.2	358.6
ME44	19	9/9	174.9	-2.6	3.2	258.3	-62.2	353.1
ME41	20	4/6	168.1	-3.5	8.7	112.5	-60.8	7.1
ME40	21	7/8	173.4	-13.0	4.4	189.9	-67.1	359.1
ME41b	21	9/9	171.3	-7.6	3.2	263.8	-63.8	2.1
ME39	22	6/6	175.5	-8.8	3.3	412.4	-65.4	353.0
ME38	23	6/6	177.6	-8.2	5.0	180.7	-65.4	348.0
ME37	24	4/4	179.1	-3.4	3.4	743.2	-63.0	344.0
ME36	26	7/7	194.3	-45.2	5.1	138.6	-77.2	257.4
ME35	27	3/5	191.4	-40.0	9.8	160.2	-78.2	279.4
ME34	28	4/5	189.7	-41.2	6.2	219.8	-80.0	279.8
ME33	29	3/8	183.7	-34.7	9.4	174.7	-79.9	321.5
ME32	30	5/5	181.1	-39.8	3.6	442.3	-83.9	332.5
ME31	31	4/4	182.8	-38.2	1.8	2739.0	-82.4	322.0

Table 2. (continued)

Flow	Unit	*n/N*	*D*, deg	*I*, deg	α_{95}	*k*	VGP Latitude, deg	VGP Longitude, deg
ME30	33	5/5	181.0	-32.7	5.5	194.4	-79.1	337.0
ME29	34	6/6	179.5	-33.1	4.2	253.1	-79.4	344.7
ME28	35	4/5	186.4	-47.8	3.4	731.8	-84.4	248.4
ME27	36	4/4	187.4	-38.8	3.8	572.8	-80.5	295.7
ME26	37	5/5	190.1	-37.8	2.1	1320.8	-78.2	289.0
ME25	38	3/5	187.4	-48.3	3.6	1176.4	-83.5	244.6
ME24	39	6/6	182.2	-46.8	4.9	187.2	-88.0	270.0
ME23	40	4/5	185.5	-36.5	5.6	268.4	-80.3	310.0
ME22	41	4/4	192.5	-42.6	4.1	500.9	-78.2	268.7
ME20	42	4/4	183.7	-37.3	2.3	1580.0	-81.5	318.0
ME21	42	4/4	188.0	-36.7	6.2	219.0	-79.0	298.8
ME19	43	7/7	181.8	-35.6	3.9	246.1	-80.9	331.3
ME18	44	4/5	201.8	-22.6	10.3	80.7	-63.6	287.4
ME17	45	3/5	190.2	-37.2	5.7	471.5	-77.8	290.2
ME16	46	2/5	193.8	-30.9	n.d.	n.d.	-72.5	292.5
ME15	47	7/8	197.6	-36.9	3.7	266.2	-72.1	275.0
ME14	48	3/8	168.0	-23.7	6.8	326.4	-70.2	19.0
ME13	49	4/4	158.2	-29.4	6.2	224.1	-66.1	44.0
ME12	50	4/8	166.0	-33.9	8.9	107.4	-73.7	36.9
ME11	51	5/5	171.6	-37.1	6.6	134.6	-79.0	27.8
ME10	52	5/5	197.0	-23.2	4.0	358.0	-67.1	294.9
ME09	53	5/5	204.2	-28.5	2.3	1098.6	-63.9	278.0
ME08	54	7/7	202.9	-24.0	3.7	272.4	-63.3	284.4
ME07	55	5/5	199.9	-23.7	4.5	290.4	-65.3	347.0
ME06	56	4/4	199.1	-26.1	5.6	271.5	-66.8	328.1
ME05	57	5/6	181.2	-33.8	5.0	234.8	-79.8	335.6
ME04	58	4/4	189.6	-35.4	8.9	108.4	-77.4	396.1
ME03	59	6/6	184.5	-15.6	7.7	75.9	-68.9	329.6
ME02	60	4/4	189.1	-44.7	9.3	98.7	-81.6	266.1
ME01	sill	2/10	192.6	-39.7	n.d.	n.d.	-77.1	277.5
			Averaged Results Within Units					
ME42/44	19	13/13	174.2	-3.5	2.6	260.7	-62.5	354.7
ME40/41b	21	16/17	172.2	-10.0	2.7	183.4	-65.2	0.9
ME20/21	42	8/8	185.8	-37.0	2.9	363.7	-80.4	307.4

under vacuum (less than 10^{-2} torr). When heating was performed under vacuum, cooling was regulated by a stream of nitrogen gas precooled by circulation within liquid nitrogen. Temperature control was provided by three thermocouples within the inner tube of the oven. In addition, accurate monitoring of the temperature of the specimens was achieved with three other thermocouples positioned within the sample holder. This adjunction was extremely helpful to duplicate pTRM checks at the same temperature (within 2-3°C) as the initial TRMs.

Special attention has been taken to detect production of chemical remanent magnetization (CRM) while inducing a pTRM. Mineralogical changes produced during heating at T_i mostly affect magnetic grains with blocking temperatures lower than T_i but could sometimes concern also grains with $T_b > T_i$. The first situation is indicated by negative pTRM checks [*Coe et al.*, 1978], while the second cannot be detected easily. Because the specimens were positioned with their *z* axis along the field direction, acquisition of CRM in such conditions should be reflected by a tendency to move the residual NRM toward the direction of the applied field [*Valet et al.*, 1996]. For this reason, attention was paid to scrutinize the demagnetization diagrams in sample and in geographic coordinates. This analysis was completed by checking the coherency of the directions obtained by alternating field and thermal demagnetization. Evidence for CRM was detected for 49 specimens (17%).

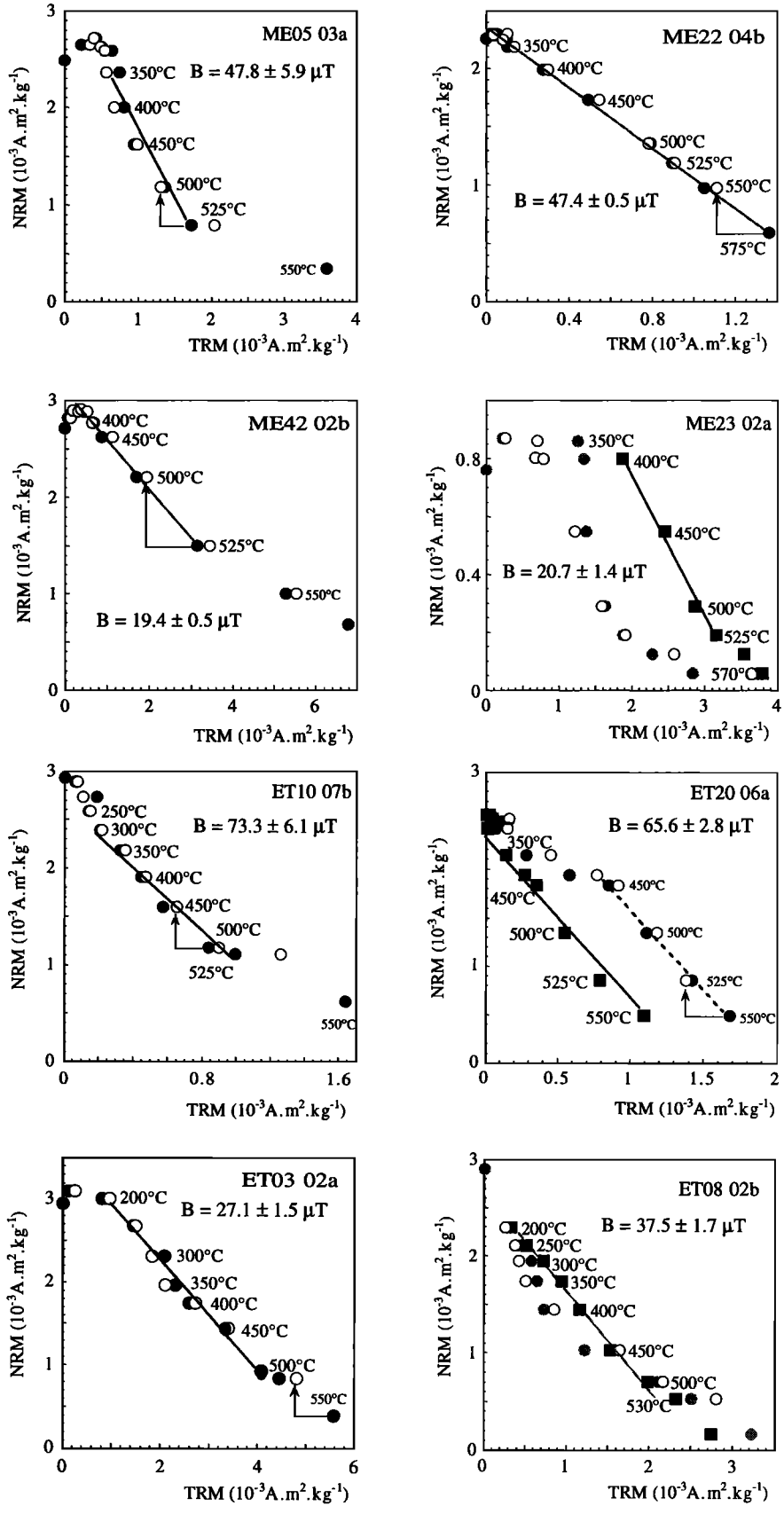


Figure 7. Typical Arai diagrams of specimens from both sections (the demagnetization diagrams of several of these specimens are shown in Figure 5). The pTRM checks (open circles) were performed systematically at all steps, one step below last heating. The NRM-TRM values are shown by solid circles. The NRM-pTRMs corrected values [*Valet et al.*, 1996] are shown by solid squares. The slopes were almost always determined above 300°. Fifteen determinations were obtained from corrected pTRMs (like ME23 02a). In a few cases such as specimen ET20 06a, the slopes, obtained before and after correction, were indistinguishable.

4.2. Characteristics of the Arai Diagrams and Corrections of the pTRMs

The paleointensity determinations were obtained from the slope of the Arai diagrams [*Nagata et al.*, 1963] which gives the ratio between the paleofield and the laboratory field. Typical NRM-TRM diagrams are shown in Figure 7. Special care was taken to interpret the Arai diagrams within the same range of temperatures as for the characteristic remanent magnetization (ChRM) particularly within high temperature intervals (up to 500°C at least and mostly 525-550°C). Segments of the NRM-TRM diagrams corresponding to blocking temperatures lower than 250-300°C were not considered because they are theoretically [*Neel*, 1949] associated with magnetic grains with relaxation times shorter than 800kyrs, i.e. younger than the age of the primary magnetization. Fifty three percent of the specimens (154) yielded straightforward paleointensity determinations and 17 flows were unsuitable for paleointensity experiments. There was no noticeable improvement when heating was performed under vacuum, except for flow ME01 which was affected by problems of viscous magnetization.

A total of 134 specimens could not be used for direct determination of absolute paleointensity. Four specimens have been affected by lightning, eight specimens (particularly from flows ME45 and ME46) carried a very strong remagnetization component, and 49 samples showed evidence of CRM acquired during the experiments. The 73 remaining specimens were characterized by nonlinear Arai diagrams, mostly with many negative pTRM checks. In these cases, we tentatively applied corrections [*Valet et al.*, 1996]. Corrections rely on the hypothesis that the loss (or gain) of TRM induced by mineralogical transformations corresponds with the difference between the pTRM check and the initial TRM. This approach requires careful documentation of the transformations by performing pTRM checks systematically after each heating step. Because transformations occurring after heating at T_{i+1} mostly affect the pTRM acquired at lower temperatures (between T_0 and T_i), the amount of magnetization that was added or removed between T_0 and T_i is indicated by the difference between the pTRM check and the initial pTRM. The quantity (pTRM-pTRMcheck)T_i is thus used for correction. Typical NRM-TRM plots with corrected pTRMs are shown in Figure 7. The corrections restored linear plots for 14 specimens (from 10 different lava flows). In these cases the corrected pTRMs were used to determine the slope of the Arai diagrams. In the other cases, the absence of linearity was incompatible with a proper determination of paleointensity and the specimens were rejected. A few diagrams (such as specimen ET20 06a in Figure 7) with one or two negative pTRM checks were characterized by high-temperature linear NRM-TRM segments. The same slopes were obtained after correction, and they were reported as uncorrected. The paleointensity values derived from the corrected diagrams are compared with the other uncorrected specimens from the same flow in Tables 3 and 4. Figure 8 plots the average corrected values versus the average uncorrected values of the other specimens from the same flows. The two determinations agree within 10 to 20%.

The average paleointensity of every flow was calculated using the weighting factor w as defined by *Prévot et al.* [1985] and is reported in Tables 3 and 4 together with the usual parameters accompanying the interpretation of the Arai diagrams. In three cases (ME01, ME05, ET18) the paleointensity determinations were limited to two samples per lava flow. Flows ME36 and ET09 were defined by a single determination and were not incorporated in the final results (which consequently includes 152 determinations instead of 154). The single determination from flow ME45 was included because of its location close to the reversal.

5. Potassium-Argon (K-Ar) Dating

Two flows (ME01 and ME15) from the ME section and two flows (ET18 and ET06) from the ET section have been dated using the refined K-Ar technique of *Cassignol and Gillot* [1982] and *Gillot and Cornette* [1986]. This technique relies on a double comparison between the argon content of the sample and the atmospheric argon, both measurements being performed under rigorously identical analytical conditions. Because of the stability of the mass spectrometer parameters and of the reproducibility of the argon isotopic measurements the detection of a fraction of radiogenic argon lower than 0.15% is made possible (see *Gillot and Cornette* [1986] for a complete description of the technique and intercomparison with other dating methods). Absolute calibration of the total ^{40}Ar signal is performed using the interlaboratory standard GL-O with an assigned value of 6.679×10^{14} at/g of ^{40}Ar*. Potassium concentrations have been obtained by flame photospectroscopy on an aliquot of the same sample groundmass fraction as that used for Ar determination (K values range from 0.9% to up to 2.1% for the groundmass of the dated flows (Table 5) and are characteristic of oceanic island basalts [*Wilson*, 1989] ranging from olivine basalts to plagioclase basalts and Hawaiite).

The flows suitable for dating were selected from their facies at the outcrop and from thin section examination. Lavas displaying any trace of alteration have been discarded. The sample preparation involved ultrasonic cleaning in dilute nitric solution followed by heavy liquid separation in order to remove the phenocrysts (olivine, pyroxene, plagioclase) in which inherited argon could be occluded [*Dalrymple and Lanphere*, 1969]. The goal was to analyze the groundmass which crystallized last during the extrusion and/or the emplacement of the lava. The dating results are reported in Table 5. Age estimates are in stratigraphic order in the ET section, but in the ME section, unit ME01 is apparently younger than ME15. This observation is not statistically significant at the 2-sigma level. However, subsequent field observations performed after the initial sampling strongly suggest that ME01 might in fact be a sill underlying ME02. As a consequence, we preferred not to incorporate the ME01 unit in the calculation of the averaged extrusion rate but rather considered this unit as a sill intrusion. The mean extrusion rate of 25 flows per 100 kyr during the Matuyama chron deduced from the ages of flow ME15 and the position of the B-M reversal at 0.78 Ma [*Bassinot et al.*, 1994] is not strikingly different from that of 19 flows per 100 kyr during the lower part of the Brunhes chron deduced from ET06 and ET18 from the ET section. It agrees with the fact that the ME section contains more flows because it is closer to the source. However, one must keep in mind that these values represent interpolation between two points and become meaningless when one takes the standard deviation into account.

Table 3. Paleointensity Results for the ET Section

Sample	Unit	ΔT, °C	f	g	q	w	B, mT	σB	Cor	$B \pm \sigma B$, mT	VADM±σ, 10^{22} A.m^{-2}	VDM±σ, 10^{22} A.m^{-2}
ET20 01a	1	450-550	0.45	0.63	14.0	9.91	56.6	1.1				
ET20 02a		200-550	0.51	0.82	14.0	5.29	69.6	2.1	C			
ET20 05a		300-525	0.54	0.77	6.00	3.01	57.3	4.0	C			
ET20 06a		450-550	0.42	0.66	6.54	4.62	65.6	2.8		61.5±5.5	12.2±1.1	11.4±1.0
ET19 01a	2	200-570	0.86	0.83	40.7	15.4	59.8	1.0				
ET19 02a		300-525	0.40	0.76	3.47	1.73	52.1	4.5				
ET19 03a		350-560	0.65	0.74	11.0	5.51	56.7	2.5				
ET19 05b		450-550	0.36	0.66	8.21	5.81	61.1	1.8		59.0±3.5	11.7±0.7	9.8±0.6
ET18 01b	3	200-450	0.38	0.72	1.46	0.84	33.7	6.4				
ET18 03a		120-300	0.26	0.62	3.41	2.41	40.3	1.9		38.6±3.3	7.7±0.7	7.1±0.6
ET17 01a		250-550	0.55	0.80	9.44	3.85	40.6	1.9				
ET17 03a		350-500	0.48	0.66	2.06	1.19	42.8	6.6				
ET17 04a		250-450	0.32	0.70	2.61	1.51	33.7	2.9		39.4±3.9	7.8±0.8	7.1±0.7
ET16 01b	4	350-550	0.49	0.73	7.46	3.73	82.5	3.9				
ET16 02b		450-550	0.34	0.66	4.32	3.06	62.9	3.3				
ET16 03b		200-525	0.43	0.78	11.2	5.00	52.2	1.6				
ET16 04a		200-500	0.28	0.66	1.60	0.80	68.4	7.9				
ET16 05b		200-525	0.50	0.80	9.26	4.14	50.9	2.2		61.4±11.6	12.2±2.3	13.8±2.6
ET15 02b	5	200-450	0.27	0.77	4.45	2.23	48.4	2.3				
ET15 03b		120-450	0.25	0.83	1.59	0.71	48.0	6.3				
ET15 05a		200-400	0.21	0.73	6.70	3.87	50.3	1.2		49.4±1.0	9.8±0.2	11.0±0.2
ET14 01c	6	200-450	0.33	0.78	2.68	1.34	52.0	5.0				
ET14 02a		200-450	0.31	0.73	1.76	1.02	51.7	6.6				
ET14 03b		300-525	0.43	0.74	4.57	2.29	49.7	3.5	C			
ET14 04a		350-500	0.32	0.64	2.70	1.91	54.3	4.1				
ET14 05a		250-450	0.26	0.66	1.74	1.00	60.9	6.7		53.0±3.9	10.5±0.8	12.0±0.9
ET13 01a	7	300-450	0.28	0.65	2.16	1.53	65.0	5.5				
ET13 01b		350-500	0.35	0.61	3.07	2.17	61.2	4.3				
ET13 02b		300-525	0.48	0.79	2.96	1.48	50.8	6.4				
ET13 05b		350-550	0.57	0.72	5.79	2.89	56.2	4.2		58.2±5.3	11.6±1.1	13.7±1.2

Table 3. (continued)

Sample	Unit	ΔT, °C	f	g	q	w	H_a μT	σH_a	Cor.	$F_a \pm \sigma F_a$ μT	VADM±σ (10^{22} A.m^2)	VDM±σ (10^{22} A.m^2)
ET10 04a	10	450-525	0.25	0.50	11.1	11.1	45.7	0.5				
ET10 05a		300-500	0.38	0.73	2.50	1.44	53.0	5.9	C			
ET10 07b		300-525	0.47	0.76	4.29	2.15	73.3	6.1				
ET10 07c		350-500	0.41	0.64	11.1	7.84	60.5	1.4		53.9±10.2	10.8±2	12.7±2.4
ET08 01a	12	300-570	0.61	0.82	10.0	4.09	29.7	1.5	C			
ET08 01c		300-450	0.30	0.62	2.43	1.72	35.8	2.7				
ET08 02b		250-500	0.53	0.78	8.93	4.47	37.5	1.7	C	34.1±3.3	6.8±0.7	7.8±0.8
ET07 01a		400-525	0.37	0.61	2.77	1.96	28.3	2.3				
ET07 01b		450-550	0.39	0.65	4.81	3.40	33.3	1.8				
ET07 02b		350-525	0.37	0.62	2.54	1.47	40.8	3.7				
ET07 03a		400-550	0.32	0.69	3.32	1.92	37.6	2.5	C			
ET07 03b		350-500	0.42	0.64	1.94	1.37	27.7	3.9		33.5±5.1	6.7±1.0	7.6±1.2
ET06 01a	13	350-560	0.80	0.73	19.6	9.82	29.6	0.9				
ET06 15		450-560	0.52	0.65	12.6	8.89	19.9	0.5				
ET06 18b		350-560	0.81	0.70	16.6	8.30	31.1	1.1		28.3±6	6.8±0.9	7.7±1.0
ET05 01a	17	350-500	0.25	0.59	1.38	0.98	9.1	0.9				
ET05 03a		450-560	0.61	0.64	10.2	7.20	4.0	0.2				
ET05 11a		500-570	0.52	0.65	4.80	3.39	9.4	0.7				
ET05 13a		500-570	0.49	0.66	8.70	6.15	9.3	0.3		7.2±2.3	1.4±0.5	1.9±0.6
ET04 01a	25	400-525	0.50	0.63	3.15	2.22	12.4	1.2				
ET04 01b		400-525	0.41	0.65	3.03	2.14	14.3	1.3	C			
ET04 02a		200-450	0.41	0.74	3.04	1.52	15.3	1.5	C			
ET04 08a		300-500	0.50	0.74	3.58	2.07	12.3	1.3	C	13.4±1.3	2.7±0.3	3.3±0.3
ET03 01b	32	200-525	0.91	0.73	48.2	19.7	36.2	0.5				
ET03 02a		200-500	0.58	0.82	8.86	3.96	27.1	1.5				
ET03 04a		400-525	0.36	0.66	4.49	3.17	59.3	3.2		37.6±13.6	7.5±2.7	8.9±3.2

ΔT, interval of temperature used for the paleointensity determination; f and g are the NRM fraction and the gap factor, respectively [*Coe et al.*, 1978]; w is the weighting factor [*Prévot et al.*, 1985]; H_a, paleointensity determination for an individual specimen; σH_a, standard error; C means that corrections [*Valet et al.*, 1996] have been applied; $F_a \pm \sigma F_a$,weighted paleointensity average for individual flow and the standard error associated with the weighted average; VADM is the virtual axial dipole moment with the corresponding standard error; VDM is the virtual dipole moment with the corresponding standard error.

Table 4. Paleointensity Results for the ME Section

Sample	Unit	ΔT °C	*f*	*g*	*q*	*w*	*B*, μT	σ*B*	Cor	*B*±σ*B*, μT	VADM±σ, 10^{22} Am^{-2}	VDM±σ, 10^{22} Am^{-2}
ME01 04	sill	200-525	0.32	0.81	5.16	2.11	28.2	1.4				
ME01 09	sill	200-560	0.71	0.65	43.2	16.3	23.7	0.3		24.2±2.3	4.8±0.5	5.1±0.5
ME02 01	60	250-525	0.48	0.81	9.05	4.05	28.9	1.2				
ME02 02a		300-525	0.56	0.77	6.73	3.36	28.2	1.8				
ME02 02b		250-525	0.48	0.79	10.3	4.61	31.0	1.1		29.5±1.2	5.9±0.2	6.0±0.2
ME03 02b	59	350-525	0.46	0.79	5.94	3.43	11.6	0.7				
ME03 04a		300-525	0.57	0.79	8.19	4.10	13.4	0.7		12.6±0.9	2.5±0.2	3.2±0.2
ME04 02a	58	300-590	0.71	0.75	14.5	5.90	22.5	0.8				
ME04 03		450-570	0.79	0.55	17.9	10.3	27.9	0.7				
ME04 04a		350-550	0.28	0.69	1.51	0.87	20.1	2.6		25.6±3.3	5.1±0.7	5.7±0.7
ME05 03a	57	400-525	0.41	0.67	2.23	1.58	47.8	5.9				
ME05 03b		400-525	0.41	0.66	1.77	1.25	46.9	7.3		47.4±0.5	9.4±0.1	10.7±1.1
ME06 01a	56	300-525	0.69	0.79	7.71	3.85	28.4	2.0				
ME06 01c		250-450	0.32	0.67	7.74	4.67	43.9	1.2				
ME06 03a		200-450	0.38	0.73	5.22	2.61	41.1	2.2		37.9±6.7	7.5±1.3	9.1±1.6
ME07 01b	55	400-525	0.52	0.64	2.69	1.90	51.4	6.3				
ME07 02a		200-500	0.44	0.77	2.55	1.28	47.1	6.3				
ME07 03a		200-500	0.31	0.75	3.75	1.88	55.7	3.5				
ME07 04a		250-500	0.41	0.74	2.89	1.45	43.0	4.5		49.9±4.7	9.9±0.9	12.1±1.1
ME08 01a	54	350-500	0.34	0.66	2.02	1.43	66.5	7.3				
ME08 01b		250-450	0.27	0.66	2.01	1.16	55.5	5.0				
ME08 01c		400-525	0.25	0.65	0.83	0.59	37.1	7.1				
ME08 03a		200-450	0.22	0.67	2.00	1.16	61.2	4.6				
ME08 04a		300-525	0.29	0.78	9.63	4.82	46.9	1.1				
ME08 05a		300-550	0.55	0.83	5.56	2.49	46.5	3.8	C	51.0±9.9	10.1±2.0	12.3±2.4
ME09 01b	53	150-550	0.56	0.87	7.29	2.58	61.7	4.1				
ME09 02a		200-500	0.48	0.72	4.67	2.33	54.0	4.0				
ME09 03a		250-500	0.48	0.74	5.07	2.53	60.3	4.2				
ME09 04		200-500	0.50	0.76	5.41	2.71	59.9	4.3		59.1±2.9	11.8±0.6	13.9±0.7
ME10 01d	52	525-575	0.49	0.36	14.5	14.5	37.3	0.5				
ME10 03a		525-575	0.41	0.34	1.96	1.96	35.3	2.6				
ME10 04a		500-560	0.69	0.23	2.26	2.26	35.3	2.4		36.9±0.9	7.3±0.2	9.0±0.2
ME11 03a	51	200-450	0.35	0.63	2.87	1.44	39.9	3.1				
ME11 04a		300-525	0.36	0.68	4.13	2.06	38.1	2.3				
ME11 05b		200-525	0.41	0.85	4.47	1.83	46.0	3.5		41.3±3.4	8.2±0.7	9.1±0.7
ME12 01a	50	350-525	0.51	0.74	2.55	1.47	24.2	3.6				
ME12 01b		400-525	0.29	0.66	1.42	1.01	39.1	5.3				
ME12 02		350-500	0.43	0.66	6.33	4.48	25.2	1.1	C	27.0±6.8	5.4±1.4	6.1±1.5

Table 4. (continued)

Sample	Unit	ΔT, °C	f	g	q	w	H_a μT	σH_a	Cor.	$F_a \pm \sigma F_a$ μT	VADM±σ, 10^{22} Am^{-2}	VDM±σ, 10^{22} Am^{-2}
ME15 01a	47	400-550	0.65	0.70	16.7	9.63	10.6	0.3				
ME15 01b		300-525	0.68	0.70	8.40	4.20	17.6	1.0				
ME15 01c		350-525	0.63	0.71	4.54	2.62	13.6	1.4				
ME15 05a		400-525	0.48	0.66	2.96	2.09	15.0	1.6		13.1±2.5	2.6±0.5	2.9±0.6
ME19 01	43	400-525	0.38	0.66	1.64	1.16	35.7	5.4				
ME19 03a		350-525	0.32	0.72	2.71	1.57	46.0	3.8				
ME19 03b		350-550	0.58	0.73	5.43	2.72	50.5	3.9				
ME19 04a		300-525	0.41	0.75	6.36	3.18	47.6	2.3				
ME19 05a		250-550	0.37	0.82	5.15	2.30	25.6	1.5		42.2±9.2	8.4±1.8	9.4±2.1
ME20 01a	42	250-500	0.22	0.75	3.26	1.63	40.3	2.0				
ME20 03a		200-525	0.35	0.78	7.34	3.00	49.5	1.8				
ME20 04a		200-525	0.32	0.79	8.66	3.88	42.8	1.3		44.7±3.9	8.9±0.8	9.8±0.9
ME21 01b		200-525	0.57	0.79	10.9	4.44	51.3	2.1				
ME21 02a		350-525	0.42	0.73	2.52	1.46	49.4	6.0				
ME21 05a		300-550	0.43	0.78	12.9	5.78	44.6	1.2		47.8±2.8	9.5±0.6	10.6±0.6
ME22 02a	41	250-525	0.43	0.78	6.85	3.06	46.1	2.3				
ME22 03a		200-525	0.48	0.78	10.7	4.77	52.6	1.8				
ME22 04b		200-575	0.72	0.84	54.2	19.2	47.4	0.5		48.2±2.8	9.6±0.6	10.1±0.6
ME23 01c	40	400-525	0.41	0.66	2.26	1.60	28.4	3.5				
ME23 02a		400-525	0.70	0.62	6.64	4.70	20.7	1.4	C			
ME23 04b		300-525	0.49	0.76	6.79	3.40	17.0	0.9		20.7±4.7	4.1±0.9	4.6±1.0
ME24 01a	39	350-550	0.49	0.79	6.17	3.08	18.9	1.2	C			
ME24 02a		450-575	0.45	0.72	5.67	3.27	14.8	0.8				
ME24 04a		400-500	0.28	0.48	11.8	11.8	17.1	0.2		17.1±1.7	3.4±0.3	3.4±0.3
ME26 01a	37	150-400	0.34	0.75	5.98	2.99	28.7	1.2	C			
ME26 02		500-560	0.34	0.46	1.65	1.65	27.1	2.6				
ME26 03a		400-550	0.49	0.74	3.68	2.12	20.9	2.0				
ME26 03b		450-560	0.59	0.62	3.96	2.08	26.1	2.4		25.9±2.9	5.1±0.6	5.7±0.6
ME27 01b	36	450-525	0.37	0.44	10.0	10.0	38.0	0.6				
ME27 02a		250-450	0.29	0.69	3.75	2.16	39.0	2.1				
ME27 04a		400-550	0.50	0.72	5.23	3.02	25.7	1.8		35.7±6.0	7.1±1.2	7.8±1.3
ME29 01a	34	250-450	0.47	0.61	14.1	8.13	21.0	0.4				
ME29 03a		200-300	0.36	0.49	3.07	3.07	19.4	1.1		20.6±0.8	4.1±0.2	4.7±0.2
ME30 01b	33	400-525	0.45	0.64	2.87	2.03	56.8	5.7				
ME30 02a		350-525	0.51	0.73	6.81	3.93	34.8	1.9				
ME30 04b		300-525	0.47	0.77	9.83	4.92	35.1	1.3				
ME30 05a		350-525	0.47	0.72	11.2	6.47	27.5	0.8		34.7±11.0	6.9±2.2	7.9±2.5

Table 4. (continued)

Sample	Unit	ΔT, °C	f	g	q	w	H_a μT	σH_a	Cor.	$F_a \pm \sigma F_a$ μT	VADM±σ, 10^{22} Am^{-2}	VDM±σ, 10^{22} Am^{-2}
ME31 01a	31	200-525	0.68	0.82	10.7	4.36	45.2	2.4				
ME31 01b		200-525	0.61	0.80	8.88	3.63	57.8	3.2				
ME31 02a		200-525	0.54	0.78	9.40	3.84	55.1	2.5		52.3±5.4	10.4±1.1	11.4±1.2
ME32 01a	30	350-550	0.50	0.79	8.67	4.33	33.2	1.5				
ME32 02a		350-500	0.35	0.64	5.26	3.72	30.3	1.3				
ME32 04a		350-570	0.76	0.82	22.6	10.1	30.9	0.9				
ME32 05a		350-570	0.76	0.82	25.5	11.4	32.3	0.8		31.7±1.1	6.3±0.2	6.8±0.2
ME38 03a	23	250-500	0.22	0.72	3.66	1.83	22.4	1.0				
ME38 04		250-525	0.34	0.78	5.03	2.25	24.7	1.3				
ME38 05b		300-525	0.42	0.72	3.34	1.67	20.9	1.9		22.9±1.6	4.6±0.3	5.9±0.4
ME39 01a	22	200-500	0.31	0.75	3.20	1.43	27.6	2.0				
ME39 01b		200-525	0.45	0.72	6.96	2.84	26.0	1.2				
ME39 03a		250-450	0.23	0.68	2.37	1.37	30.6	2.0		27.5±1.9	5.5±0.4	7.0±0.5
ME40 1a	21	400-525	0.42	0.65	1.91	1.35	13.6	1.9				
ME40 02a		400-525	0.42	0.66	3.39	2.40	8.5	0.7				
ME40 02b		400-525	0.43	0.66	7.20	5.10	8.6	0.3				
ME40 04a		400-525	0.44	0.65	7.76	5.49	13.7	0.5				
ME40 04b		400-525	0.44	0.65	5.08	3.59	13.9	0.7		11.6±2.5	2.3±0.5	2.9±0.6
ME41b 05a		350-550	0.70	0.75	17.0	8.47	9.2	0.3				
ME41b 05b		350-525	0.41	0.63	11.2	6.48	13.8	0.3				
ME41b 11a		500-575	0.22	0.60	0.61	0.43	16.0	3.4		11.3±2.8	2.2±0.6	2.9±0.7
ME42 01a	19	350-525	0.43	0.65	7.06	4.08	16.3	0.7				
ME42 02a		350-550	0.69	0.73	8.00	8.00	14.3	0.5				
ME42 02b		350-525	0.45	0.63	6.12	6.12	19.4	0.5		16.5±2.1	3.3±0.4	4.2±0.5
ME44 01a		350-500	0.20	0.59	7.73	5.47	19.4	0.3				
ME44 01b		400-550	0.68	0.71	7.53	4.35	17.8	1.1				
ME44 05a		350-575	0.88	0.77	13.5	6.02	12.3	0.6				
ME44 06		400-525	0.42	0.59	4.63	3.27	10.7	0.6		15.3±3.6	3.0±0.7	4.0±0.9
ME45a 04b	18	500-550	0.23	0.50	0.48	0.48	7.6	1.8		7.6±1.8	1.5±0.4	2.0±0.5
ME47 09a	16	200-525	0.80	0.80	55.7	22.7	33.2	0.4				
ME47 11		400-560	0.90	0.59	12.0	6.90	41.5	1.9				
ME47 12a		250-575	0.83	0.85	35.6	13.44	38.2	0.8		36.1±3.4	7.1±0.7	8.3±0.7
ME48 02a	15	400-550	0.47	0.69	15.3	8.81	48.5	1.0				
ME48 02b		400-550	0.39	0.74	26.6	15.4	44.3	0.5				
ME48 04		350-550	0.56	0.79	23.5	11.7	42.7	0.8		44.8±2.4	8.9±0.5	10.4±0.6
ME49 01	14	400-525	0.46	0.66	5.62	4.00	30.3	1.6				
ME49 02		350-525	0.66	0.74	4.04	2.34	37.4	4.5		32.9±3.6	6.5±0.7	7.7±0.8

Same legend as in Table 3.

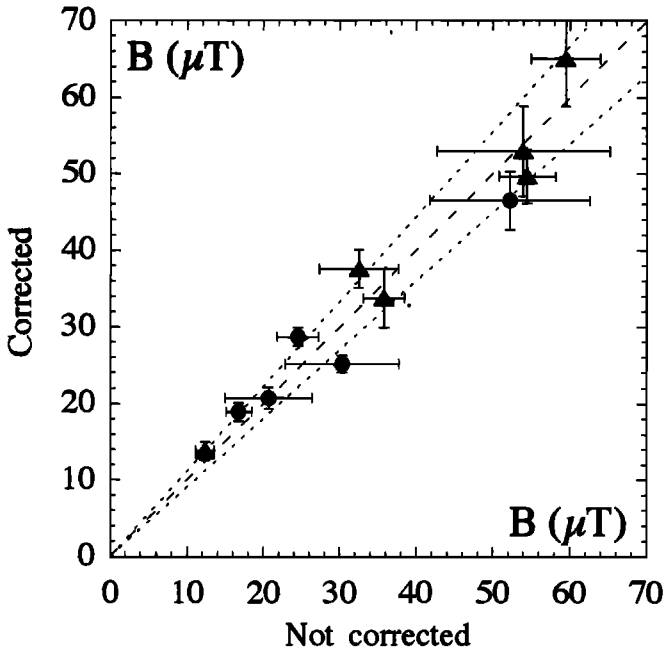


Figure 8. Mean paleointensity values calculated from the corrected NRM-TRM diagrams versus the mean values from the uncorrected diagrams.

6. Paleomagnetic Results and Time Independent Characteristics

6.1. Correlation of the Two Sections

The correlation between the two sections depends on the position of the Brunhes-Matuyama reversal which was identified between flows ET04 and ET05 in the ET section and between flows ME46 and ME47 in the ME section. The lower flows of the ET section (ET 01-04) are characterized by inclinations between -43.5° and -28.3°. These values differ from the much shallower inclinations of flows ME37 to ME45 which precede the reversal and which must be temporally very close to it. Thus this immediately pretransitional period is not represented in the ET section (which is further away from the eruptive center). The best way to correlate flows ET 01 to ET 05 with the ME section was to search for the best concordance between paleomagnetic directions, paleointensity and rock magnetic parameters. We assumed that units that have been erupted at the same period have similar directions and paleointensities. The ARM/K grain size ratio was used also as an indicator of similar rock magnetic properties. As a result, flows ET01 and ET02 were correlated with flows ME28 and ME30, respectively. These groups of two flows have been assigned unit numbers but the directions of the flows have not been averaged together. Flow ET03 was inserted between ME30 and ME31. Flow ET04, with a low reverse

Table 5. K-Ar Ages of the Four Dated Lava Flows

Sample	K%	% $^{40}Ar^*$	$^{40}Ar^*$, 10^{11}at./g	Age, Ma	Uncertainty, 1σ, Ma
ET56	1.722	15.08	12.693	0.706	0.012
		14.21	12.510	0.695	0.012
		19.32	12.798	0.711	0.011
Average				0.705	0.012
ET65b	2.061	15.81	15.940	0.740	0.013
		19.22	16.179	0.751	0.012
		15.40	16.360	0.760	0.013
Average				0.751	0.013
ME15	1.523	9.72	14.416	0.906	0.019
		3.02	14.417	0.906	0.047
Average				0.906	0.026
ME01	0.934	13.98	8.4399	0.865	0.015
		11.84	8.3178	0.852	0.016
		9.26	8.2722	0.848	0.018
Average				0.856	0.017

inclination, which immediately precedes the transition was inserted between flows ME36 and ME37. Flow ET05, with low negative inclination and normal declination, is even closer to the transition and was positioned between flows ME45 (reverse polarity) and ME 47 (normal polarity). The other flows from the ET section are associated with the younger part of the sequence based on the results of dating.

Five pairs of lava flows (see Tables 1 and 2) with nearly identical paleomagnetic results and rock magnetic properties have been identified and the results averaged, assuming they represent single units. Accordingly, the succession of the two sequences has been defined by unit numbers from the most recent part at the top of the ET section (unit 01 is associated with flow ET20) to the oldest one at the bottom of the ME section (unit 60 for flow ME02). No unit number was assigned to the sill ME01, which has been plotted by referring directly to its K-Ar age. The correspondence between units and flows is given in Tables 1 to 4.

6.2. Directions

In Figure 9a and 9b the mean declinations and inclinations of each lava flow have been plotted as a function of unit number. After averaging the directions within each unit, we calculated the latitudes of the virtual geomagnetic pole (VGP) plotted in Figure 9c. A first feature evident from these plots is the absence

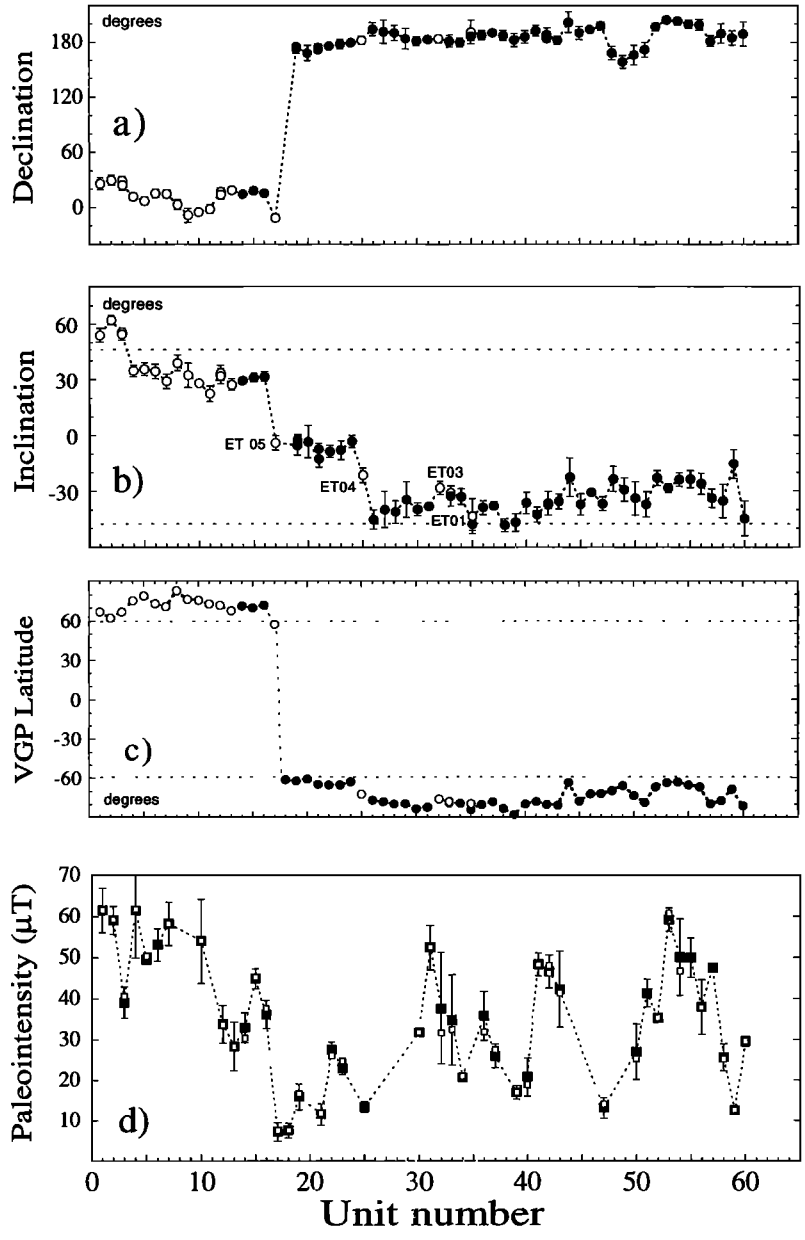


Figure 9. (a) mean declination and (b) inclination of each individual lava flow as a function of their unit number. (c) Mean VGP latitude obtained for each unit (solid circles, ME section; open circles, ET section). (d) The complete data set (solid symbols) is compared with the subset of determinations obtained with a quality factor higher than 5 (open symbols). There is negligible difference between the two sets of results.The sill ME01 was not numbered as a unit and is thus not shown in these results.

of transitional declinations, whereas the inclination seems to change more gradually. Units 17 to 24 exhibit inclinations close to 0°. However, the declinations do not significantly deviate from the north-south geographic axis, so that the VGP latitudes remain close to 60° and cannot be strictly considered as transitional. Although they cover only the past 500 years, studies of historical secular variations [*Soler et al.*, 1984] at La Palma have shown that the inclinations have not deviated by more than 27° from the inclination (47°) of the geocentric axial dipole (GAD). Note that units 17 to 24 with inclinations as far as 45° away from the GAD, and VGP latitudes close to 60°, lie far beyond these recent limits. The low paleointensity values (between 7 and 23 μT) associated with these units indicate a substantial decrease of the dipole field. Thus, whether or not they can be seen as "transitional, " these directions are likely to be the result of an increase of the nondipole components with respect to the dipole component of the field.

The mean declination of 184±5° calculated from 49 lava flows with reverse polarity is not perfectly opposite to the value of 11±9° obtained from 19 flows with normal polarity. It is not impossible that the eastward declinations could reflect a small rotation of the site, caused by faulting associated with the formation of the caldera. However, the fact that the reverse and normal inclinations averages of -30±4° and 35±7°, respectively, are both significantly lower than the 47.4° inclination at the site due to the geocentric axial dipole (GAD) does not favor the hypothesis of tilting. In any case the overlapping circles of confidence of the normal and reverse Fisherian averages indicate the absence of significant asymmetry between the two polarities.

6.3. Paleointensity

Figure 9d shows the mean paleointensity values calculated for each successive unit, with the standard errors associated with the weighted average. Records of absolute paleointensity often show relatively large within flow dispersion. In the present case, the mean dispersion is 4.2 μT, which represents roughly 10% of the average paleointensity. Eleven determinations have been obtained from three points, but most of them are associated with a value of the f parameter [*Coe et al.*, 1978] larger than 0.4. Several approaches have been used to explore the robustness and the stability of the results. We selected determinations based on five-point NRM-TRM segments with f>0.25, four-points NRM-TRM segments with f>0.3 and three-point segments with f>0.4. The results changed by less than 10%. Another approach was to rely on paleointensity determinations with a quality factor higher than 5. The paleointensity record emerging from this selection has a mean quality factor of 12 and is shown by open squares in Figure 9d. The variations duplicate those of the total data set.

The first characteristic emerging from Figure 9d is the difference between the mean prereversal and postreversal field intensities. The mean reverse and normal paleointensities are 31±13μT (N=31) and 47±12μT (N=13), respectively. Low pretransitional paleointensity values have been reported for the same reversal recorded on the eastern side of the island at Los Sauces [*Quidelleur and Valet*, 1996]. Strong posttransitional field values have been reported in a Pliocene reversal recorded at Kauai [*Bogue and Paul*, 1993], in the upper Jaramillo subchron recorded from Tahiti [*Chauvin et al.*, 1990] and can be seen also after the 15 Ma old Steens Mountain reversal [*Prévot et al.*, 1985].

7. Discussion

7.1 Time Dependent Characteristics

It is clear that chronology is essential for a thorough interpretation of the record. Based on the ages estimates obtained above, we assigned an age to each unit by assuming a constant eruption rate between the dated lava flows. This approach is not perfect but remains the only way to obtain a reasonable chronology. The mean eruption rate of the ME sequence was calculated from the ages of flow ME15 (Table 5) and the position of the Brunhes-Matuyama (780 ka) boundary [*Bassinot et al.*, 1994; *Singer and Pringle*, 1996] between flows ME46 and ME47 (unit 16). The ages of flows ME47 to ME49 were calculated by extrapolation, using the same eruption rate. The ages of the ET section have been interpolated from the ages of flows ET 06 and ET 18 (Table 5). The virtual dipole moments (VDMs) have been plotted as a function of time in Figure 10a. The dated flows can be identified by the presence of horizontal error bars. Note that the lengths of the error bars allow flexibility in this chronology. Keeping in mind these uncertainties, Figure 10a depicts different patterns for the pretransitional and posttransitional phases. The overall tendency of the prereversal phase rather indicates a slow decrease of the field intensity before the M-B reversal while there is a relatively rapid postreversal recovery.

7.2. Integration of Paleointensity Signals from Sedimentary and Volcanic Records

Unfortunately, there are no other detailed dated records of absolute paleointensity with which the present one can be compared. The study [*Quidelleur and Valet*, 1996] performed at Los Sauces, in the same island, included 13 paleointensity determinations before the reversal. Both records are characterized by low pretransitional inclinations (with similar VGP positions) and weak paleointensities. The posttransitional period at Los Sauces shows two flows with normal polarity and high paleointensity values that could tentatively be also compared with the present data set. However, the absence of chronology at Los Sauces makes any direct comparison very premature. Indeed, the upper part of the Los Sauces sequence is characterized by low VGP latitudes which were initially interpreted as an excursion. Recent K/Ar determinations [*Quidelleur et al.*, 1997] confirm this interpretation and show that this episode is, in fact, much younger than the youngest flows of the present study. Two other volcanic records of the Brunhes-Matuyama reversal have been published. Unfortunately, the first one from Tahiti [*Chauvin et al.*, 1990] has no chronology and the second one [*Brown et al.*, 1994] does not include paleointensity determinations.

The present succession of paleointensity variations may also be compared with records of relative paleointensity from sediments. The record from Ocean Drilling Program (ODP) site 851 in the equatorial Pacific [*Valet and Meynadier*, 1993] was selected for that purpose because this is the most complete and accurately dated record including the B-M reversal (the sediments were dated by tuning the density variations [*Shackleton et al.*, 1995] to the earth's orbital parameters and subsequently confirmed by $\delta^{18}O$ stratigraphy). The measurement interval at ODP site 851 (typically 2 to 4cm) and the mean deposition rate (2 cm/kyr) mean that the data points represent average over at least 1 kyr. This value increases significantly if one incorporates smoothing associated with the

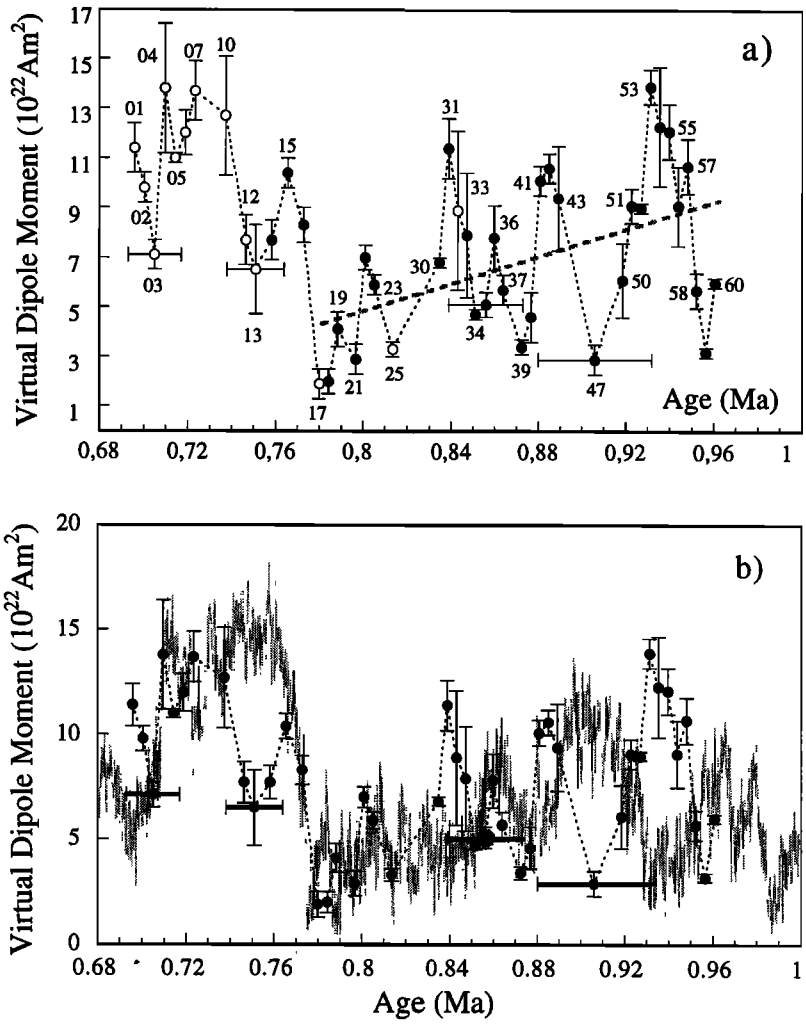


Figure 10. (a) Virtual dipole moments (VDMs) of the successive volcanic units as a function of age. Horizontal error bars indicate the error on the K-Ar ages. The vertical bars give the standard deviation about the mean paleointensity. The dashed line through the Matuyama chron was calculated by linear regression and indicates the overall tendency (R=0.5) of the field intensity decrease. The position of the sill ME01 (shaded dot) was plotted by referring to its K/Ar age. (b) Comparison between the VDMs derived from ODP site 851 (shaded line) (see text for explanation) and the absolute volcanic VDMs.

recording process. Thus, based on recent determinations of the time constants of the non dipole components [*Hulot and Le Mouël*, 1994], the relative paleointensities from the ODP site 851 sediments are regarded as estimates of relative Virtual axial dipole moments (VADMs). In contrast, volcanic records provide instantaneous readings of the total local field which are interpreted directly in terms of VDMs. The two data sets could be reconciled by averaging successive volcanic VDMs over finite periods of time but there is no indication that the temporal distribution of the flows provides a comprehensive statistical sampling of the nondipole field over short time intervals. These problems have been addressed by simulating the variations of the total field intensity at the site. The VADMs derived from the sedimentary record at ODP site 851 were taken as reflections of the variations of the axial dipole. The sign of the dipole was assigned by the polarity of magnetization in the sediment. The strength of the dipole was deduced after calibrating the mean value of relative paleointensity with the mean value of the present record of absolute paleointensity. The variations linked to the equatorial dipole and to the nondipole fields were generated by assuming that the characteristic times, means, and variances of the Gauss coefficients in spherical harmonic analysis were the same as those defined for the recent period (the past 2000 years) by *Hongre et al.* [1998]. The calculations of these statistical parameters, which characterize the present field as a stochastic process, are identical to those described by *Hulot and Le Mouël* [1994].

In Figure 10b the volcanic VDMs are compared with the sedimentary VDMs as generated by the calculation described above. The two records exhibit similar patterns but the curves do not coincide within several time intervals. They could be reconciled by taking into account the uncertainties in the ages, of the order of 20-25 ka (i.e. typically the wavelength of the dominant paleointensity features) and thus large enough to shift

features with respect to each other. Thus, in the absence of extremely detailed chronology it is not possible to derive an exact match between the two curves.

8. Conclusion

This study provides the most detailed record of absolute paleointensity obtained so far for the period surrounding the Brunhes-Matuyama reversal. The correlation of the two sections that were sampled for this purpose suffers little uncertainties and provides a coherent and detailed documentation of field behavior over a 300-kyr-long period encompassing the last reversal. There are no actual transitional directions but several shallow inclinations yielding VGP latitudes around 60°. Fifty-five percent of the specimens (158) yielded reliable paleointensity determinations, 15 of which were obtained after correction. A first characteristic of the record is the existence of high field intensities after the transition, which confirms previous observations from other detailed records of absolute paleointensity surrounding other reversals. Large fluctuations, with values between 7 and 59 μT seem to be superimposed on an overall long-term decrease of the field during the period that precedes the transition. A direct consequence is the existence of low pretransitional paleointensities which are in contrast with the postreversal phase. However, a more detailed interpretation of the pattern of the successive field changes is limited by the error bars on the ages. This unavoidable uncertainty imposes flexibility to the correlation with the records of relative paleointensity which may or may not match with the volcanic data set. Such flexibility must be considered as a limitation for studies of short-term geomagnetic variations from volcanic sequences. We conclude that this first volcanic record provides some support but cannot prove the existence of asymmetrical saw-toothed variations of the geomagnetic field intensity across the last reversal. In other words, this leaves the debate opened to some extent.

Acknowledgments. The authors are pleased to thank Isabelle LeMeur, Yohan Guyodo, Jonathan Glen, Anna Montagnini and Serge Correia for their help in sampling and also at several steps of the study. Maxime LeGoff and Emmanuel Tric provided technical assisitance. This is IPGP contribution 1571.

References

Abdel-Monem, A., N.D. Watkins and P.W. Gast, Potassium-argon ages, volcanic stratigraphy, and geomagnetic polarity history of the Canary Islands: Tenerife, La Palma and Hierro, *Am. J. Sci., 272*, 805-825, 1972.

Ancochea, E., F. Hernán, A. Cendrero, J.M. Cantagrel, J.M. Fúster, E. Ibarrola and J. Coello, Constructive and destructive episodes in the building of a young Oceanic Island, La Palma, Canary Islands, and genesis of the Caldera de Taburiente, *J. Volcanol. Geotherm. Res., 60*, 243-262, 1994.

Baksi, A.K., V. Hsu, M.O. McWilliams, and E. Farrar, $^{40}Ar/^{39}Ar$ dating of the Brunhes-Matuyama geomagnetic field reversal, *Science, 256*, 356-357, 1992.

Bassinot, F., L.D. Labeyrie, E. Vincent, X. Quidelleur, N.J. Shackleton and Y. Lancelot, The astronomical theory of climate and the age of the Brunhes-Matuyama reversal, *Earth Planet. Sci. Lett., 126*, 91-108, 1994.

Bogue, S.W., and H.A. Paul, Distinctive field behavior following geomagnetic reversals, *Geophys. Res. Lett., 20*, 2399-2402, 1993.

Brown, L., J. Pickens, and B. Singer, Matuyama-Brunhes transition recorded in lava flows of the Chilean Andes: Evidence for dipolar fields during reversals, Geology, *22*, 299-302, 1994.

Cassignol, P., and P.Y. Gillot, Range and effectiveness of unspiked potassium-argon dating: Experimental ground-work and applications, in *Numerical Ddating in Stratigraphy*, edited by G.S. Odin, pp. 159-179, J. Wiley, New York, 1982.

Chauvin, A., P. Roperch, and R.A. Duncan, Records of geomagnetic reversals from volcanic islands of French Polynesia, 2, Paleomagnetic study of a flow sequence (1.2-0.6 Ma) from the island of Tahiti and discussion of reversal models, *J. Geophys. Res., 95*, 2727-2752, 1990.

Coe, R.S., Paleointensities of the Earth's magnetic field determined from Tertiary and Quaternary rocks, *J. Geophys. Res., 72*, 3247-3262, 1967.

Coe, R.S., C.S. Grommé, and E. Mankinen, Geomagnetic paleointensities from radiocarbon dated lava flows on Hawaii and the question of the Pacific nondipole low, *J. Geophys. Res., 83*, 1740-1756, 1978.

Dalrymple, G.B., and M.A. Lanphere, *Potassium-Argon Dating*, 258pp., W.H. Freeman, New York, 1969.

Dash, B.P., and E. Bosshard, Seismic and gravity investigations around the western Canary Islands, *Earth Planet. Sci. Lett., 7*, 169-177, 1969.

Day, R., M. Fuller, and V.A. Schmidt, Hysteresis properties of titanomagnetites: Grain-size and compositional dependence, *Phys. Earth Planet. Inter., 13*, 260-267, 1977.

Fisher, R.A., Dispersion on a sphere, *Proc. R. Soc. London, ser. A, 217*, 295-305, 1953.

Frank, M., B. Schwarz, S. Baumann, A. Mangini, and P.W. Kubik, Variations of geomagnetic field intensity and cosmogenic radionucleide production rate during the last 200 kyr from global stacked ^{10}Be deep-sea sediments records, *Earth Planet. Sci. Lett., 146*, 121-130, 1997.

Fúster, J.M., E. Ancochea, E. Ibarrola, J. Coello, F. Hernán, J.M. Cantagrel, C. Jamond, and A. Cendrero, Ensayo de cronoestratigrafía de la Isla de La Palma, paper presented at *Ténérife ESF Meeting on Canarian Volcanism*, Eur. Space Found., , 1989.

Gillot, P.Y., and Y. Cornette, The Cassignol technique for potassium-argon dating, precision and accuracy: Example from the late Pleistocene to recent volcanics from southern Italy, *Chem. Geol., 59*, 205-222, 1986.

Guyodo, Y., and J.-P. Valet, Relative variations in geomagnetic intensity from sedimentary records: The past 200,000 years, *Earth Planet. Sci. Lett., 143*, 23-36, 1996.

Hongre, L., G. Hulot and A. Kohkhlov, An analysis of the geomagnetic field over the past 2000 years, *Phys. Erath Planet. Inter.*, 106, 311-335, 1998.

Hulot, G., and J.L. Le Mouël, A statistical approach to the Earth's main magnetic field, *Phys. Earth Planet. Inter., 82*, 167-183, 1994.

King, J.W., S.K. Banerjee, J. Marvin, and Ö. Özdemir, A comparison of different magnetic methods for determining the relative grain size of magnetite in natural materials: Some results from lake sediments, *Earth Planet. Sci. Lett., 59*, 404-419, 1982.

King, J.W., S.K. Banerjee, and J. Marvin, A new rock-magnetic approach to selecting sediments for geomagnetic paleointensity studies: Application to paleointensity for the last 4000 years, *J. Geophys. Res., 88*, 5911-5921, 1983.

Kok, Y.S., and L. Tauxe, Saw-tooth pattern of relative paleointensity records and cumulative viscous remanence, *Earth Planet. Sci. Lett., 137*, 95-99, 1996.

Laj, C., C. Kissel, F. Garnier, and E. Herrero-Bervera, Relative geomagnetic field intensity and reversals for the last 1.8 Ma from a central equatorial Pacific core, *Geophys. Res. Lett., 23*, 3393-3396, 1996.

Mazaud, A., Saw-tooth variation in magnetic intensity profiles and delayed acquisition of magnetization in deep sea cores, *Earth Planet. Sci. Lett., 139*, 379-386, 1996.

Meynadier, L., and J.-P. Valet, Relative paleointensity in sediments: what should we believe?, paper presented at IUGG XXI General Assembly, Boulder, Colo., 1995.

Meynadier, L., and J.-P. Valet, Post-depositional realignment of magnetic grains and asymmetrical saw-tooth patterns of magnetization intensity, *Earth Planet. Sci. Lett., 140*, 123-132, 1996.

Meynadier, L., J.-P. Valet, F.C. Bassinot, N.J. Shackleton, and Y. Guyodo, Asymmetrical saw-tooth pattern of the geomagnetic field intensity from equatorial sediments in the Pacific and Indian Oceans, *Earth. Planet. Sci. Lett., 126*, 109-127, 1994.

Meynadier, L., J.-P. Valet, and N.J. Shackleton, Relative geomagnetic intensity during the last 4 m.y. from the equatorial Pacific, , *Proc Ocean Drilling Program., Sci. Results, 138.*, 779-795, 1995.

Meynadier, L., J.-P. Valet, Y. Guyodo, and C. Richter, Saw-toothed variations of relative paleointensity and cumulative viscous remanence : Testing the records and the model, *J. Geophys. Res., 103*, 7095-7105, 1998.

Nagata, T., Y. Arai, and K. Momose, Secular variation of the geomagnetic total force during the last 5000 years, *J. Geophys. Res.*, *68*, 5277-5281, 1963.

Navarro, J.M., and J. Coello, Geological map of the National Park of Caldera de Taburiente, Instituto Nacional para la Conservacion de la Naturaleza, La Palma, 1993.

Néel, L., Théorie du trainage magnétique des férromagnétiques aux grains fins avec applications aux terres cuites, *Ann. Geophys.*, *5*, 99-136, 1949.

Prévot, M., E.A., Mankinen, R.S. Coe, and C.S. Grommé, The Steens Mountain (Oregon) geomagnetic polarity transition, 2, Field intensity variations and discussion of reversal models, *J. Geophys. Res.*, *90*, 10417-10448, 1985.

Quidelleur, X., and J.-P. Valet, Geomagnetic changes across the last reversal recorded in lava flows from La Palma, Canary Islands, *J. Geophys. Res.*, *101*, 13755-13773, 1996.

Quidelleur, X., P.Y., Gillot, and J., Carlut, Link between excursions and paleointensity inferred from abnormal field directions recorded at La Palma around 600Ka, *EosTrans. AGU, 78, Fall Meet. Suppl.*, F181, 1997.

Shackleton, N. J., S. Crowhurst, T. Hagelberg, N. G. Pisias, and D. A. Schneider, A new late Neogene time scale: application to Leg 138 sites, *Proc. Ocean Drilling Program, Sci. Results, 138,* 503-514, 1995.

Singer, B.S., and M.S. Pringle, Age and duration of the Matuyama-Brunhes geomagnetic polarity reversal from $^{40}Ar/^{39}Ar$ incremental heating analyses of lavas, *Earth Planet. Sci. Lett.*, *139*, 47-61, 1996.

Soler, V., J.C. Carracedo and F. Heller, Geomagnetic secular variation in historical lavas from the Canary Islands, *Geophys. J. R. Astron. Soc.*, *78*, 313-318, 1984.

Thellier, E., and O. Thellier, Recherches géomagnétiques sur des coulées volcaniques d'Auvergne, *Ann. Geophys.*, *1*, 37-52, 1944.

Thibal, J., J.P. Pozzi, V. Barthès and G. Dubuisson, Continuous record of geomagnetic field intensity between 4.7 and 2.7 Ma from downhole measurements, *Earth Planet. Sci. Lett.*, *136*, 541-550, 1995.

Valet, J.P., and L. Meynadier, Geomagnetic field intensity and reversals during the past four million years, *Nature*, *366*, 91-95, 1993.

Valet, J.P., J. Brassart, I. Le Meur, V. Soler, X. Quidelleur, E. Tric, and P.Y. Gillot, Absolute paleointensity and magnetomineralogical changes, *J. Geophys. Res.*, *101*, 25029-25044, 1996.

Valet, J.P., T. Kidane, V. Soler, J. Brassart, V. Courtillot and L. Meynadier, Remagnetization in lava flows recording pretransitional directions, *J. Geophys. Res.*, *103*, 9755-9775, 1998.

Verosub, K.L., E. Herrero-Bervera, and A.P. Roberts, Relative geomagnetic paleointensity across the Jaramillo subchron and the Matuyama/Brunhes boundary, *Geophys. Res. Lett.*, *23*, 467-470, 1996.

Wilson, M., *Igneous Petrogenesis: A Global Tectonic Approach*, 466 pp., Chapman and Hall, New York, 1989.

J. Bassart, L. Hongre, and J.-P. Valet, Institut de Physique du Globe de Paris, Laboratoire de Paléomagnétisme, UMR CNRS 7577, 4 Place Jussieu, 75252 Paris Cedex 05, France.(valet@ipgp.jussieu.fr)

P.-Y. Gillot and X. Quidelleur, Laboratoire de Géochronologie, Université Paris XI, Orsay, France.

V. Soler, Instituto de productos Naturales y Agrobiologia de Canarias, Estacion Volcanologica de Canarias, La Laguna, Tenerife, Spain.

(Received December 2, 1997 ; revised November 3, 1998 ;
accepted November 12, 1998.)